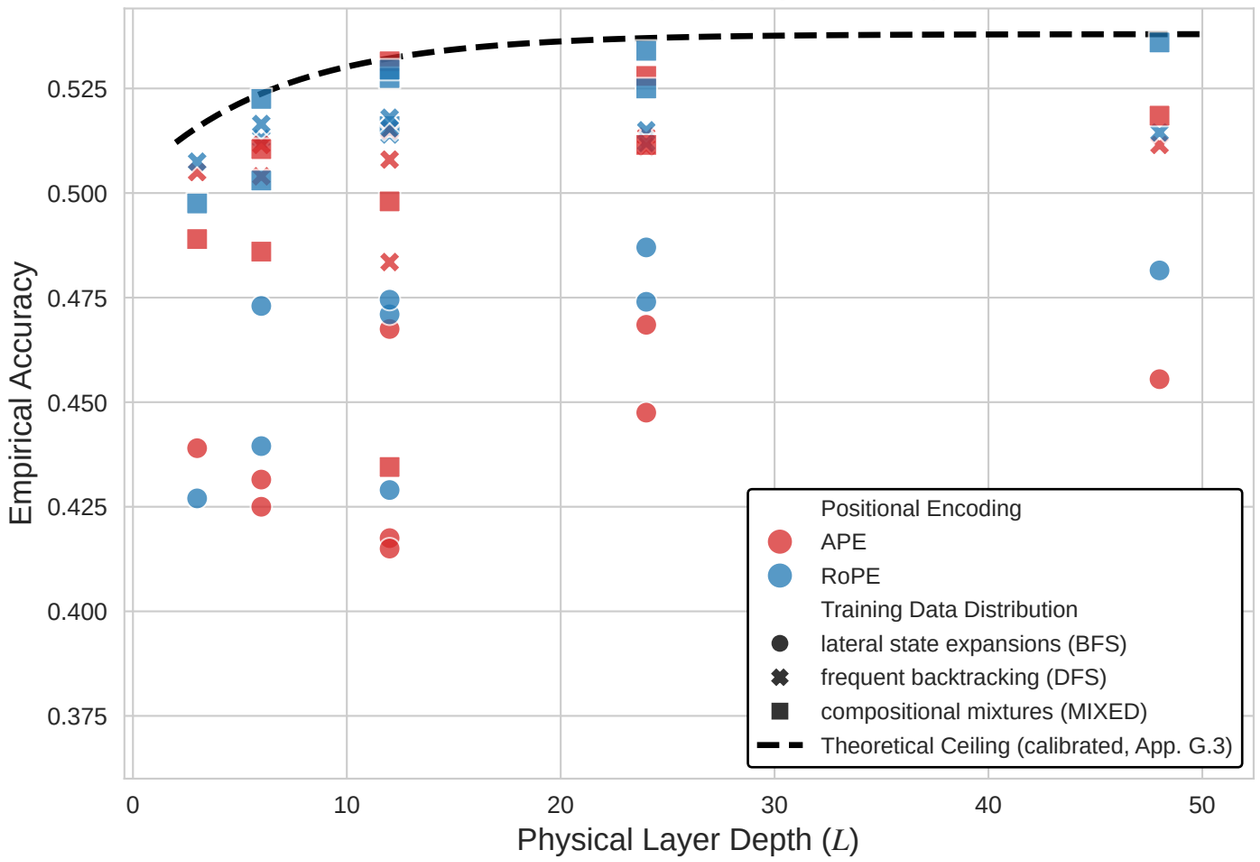
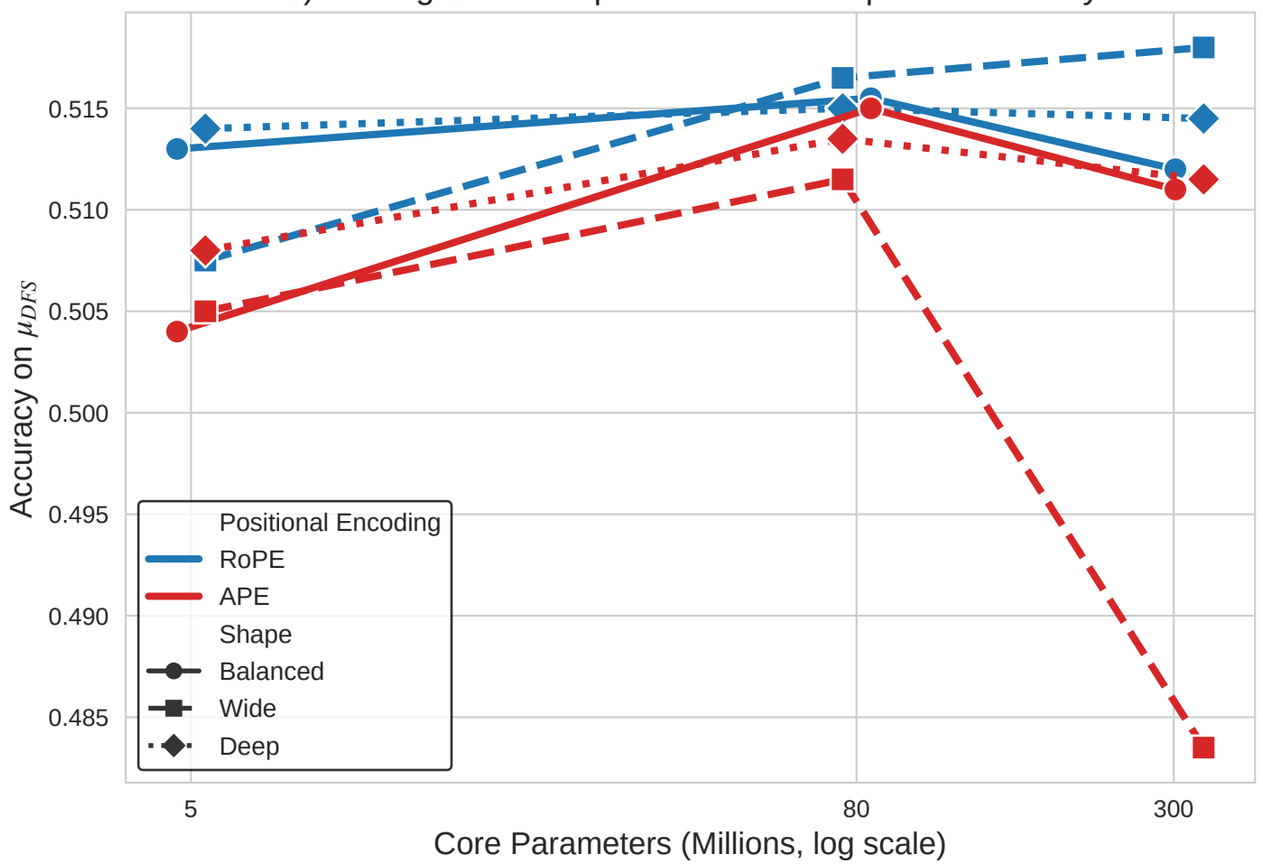
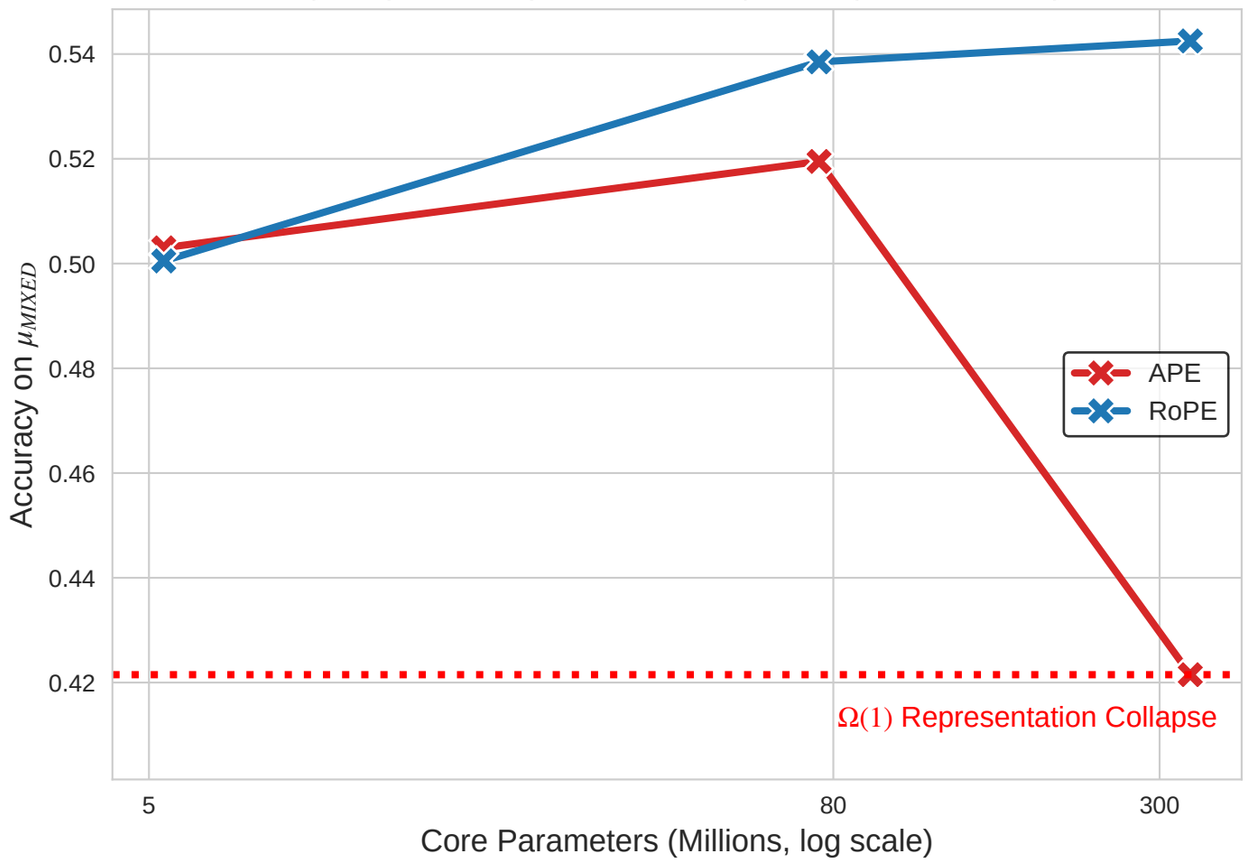
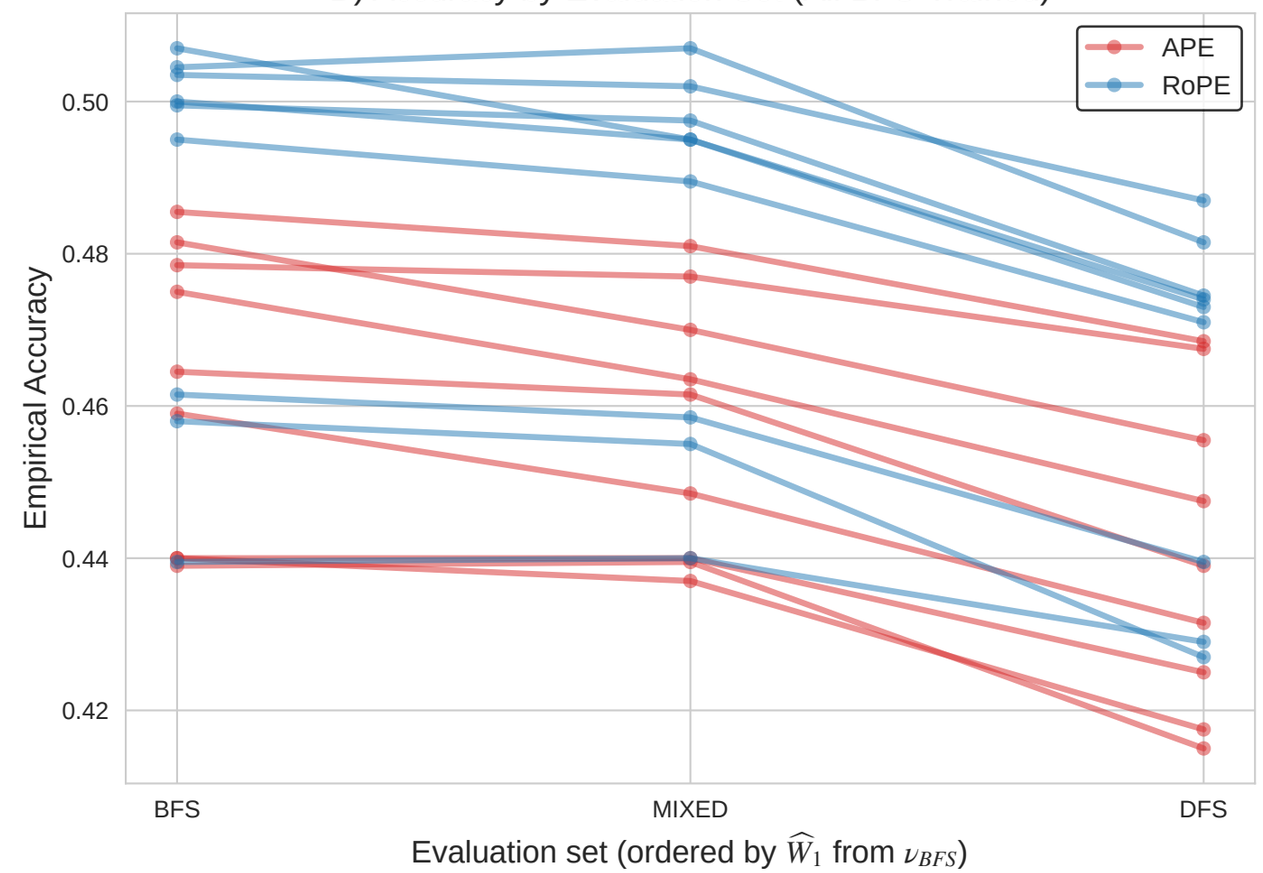

A) The Supremum Capacity Limit on μ_{DFS}
Empirical Accuracy
0.525
0.500
0.475
0.450
0.425
0.400
0.375
0
10
20
30
40
50
Physical Layer Depth (L)
Positional Encoding
APE
RoPE
Training Data Distribution
lateral state expansions (BFS)
frequent backtracking (DFS)
compositional mixtures (MIXED)
Theoretical Ceiling (calibrated, App. G.3)


B) Scaling Limits: Representation Collapse vs. Stability
Accuracy on μ_{DFS}
0.515
0.510
0.505
0.500
0.495
0.490
0.485
5
80
300
Core Parameters (Millions, log scale)
Positional Encoding
RoPE
APE
Shape
Balanced
Wide
Deep


C) Sequence Equivariance on μ_{MIXED} (Wide Models)
Accuracy on μ_{MIXED}
0.54
0.52
0.50
0.48
0.46
0.44
0.42
5
80
300
Core Parameters (Millions, log scale)
APE
RoPE
$\Omega(1)$ Representation Collapse


D) Accuracy by Evaluation Set (All BFS-Trained)
Empirical Accuracy
0.50
0.48
0.46
0.44
0.42
BFS
MIXED
DFS
Evaluation set (ordered by $\widehat{W}_1$ from ν_{BFS})
APE
RoPE